Ansh Chandrakar

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LinkedIn Profile Portfolio Website

ABOUT ME

MSc Physics student at IIT Hyderabad with a strong foundation in computational physics. Passionate about exploring astrophysics and high-energy phenomena through numerical modeling and simulation. Experienced in utilizing Python to bridge theoretical physics with computational reality.

EDUCATION

Master of Science in Physics 2025 - Present
Indian Institute of Technology, Hyderabad, Telangana
- *CGPA*: 6.71/10

Bachelor of Science (Honours) in Physics 2022 - 2025
Birla Institute of Technology, Mesra, Ranchi, Jharkhand
- *CGPA*: 8.6/10
- *Elective Courses*: Computational Physics Skills, Radiation Safety, Nuclear and Particle Physics, Classical Dynamics, Astronomy and Astrophysics

High School 2017 - 2021
Sri Sankara Vidyalaya, Bhilai, Chhattisgarh
- *12th*: 91.8%
- *10th*: 92.4%

RESEARCH PROJECTS

MSc Winter Project: Non-Reciprocal Ising Model with Stochastic Resetting
- Studied non-equilibrium statistical mechanics of Ising systems with asymmetric (non-reciprocal) interactions that violate detailed balance.
- Analyzed stochastic spin-flip dynamics and the emergence of non-equilibrium steady states with probability currents.
- Investigated the effects of stochastic resetting on system dynamics and steady states, highlighting the interplay between non-reciprocity and resetting.

Bachelor's Dissertation: Entanglement in Coupled Quantum Oscillators Using QuTiP
- Explored entanglement in coupled quantum harmonic oscillators using the Quantum Toolbox in Python (QuTiP) as part of the Bachelor's dissertation project.
- Investigated coupling mechanisms, unitary evolution, and computed entanglement measures such as Von Neumann entropy and logarithmic negativity across various systems.
- Extended the study to hybrid systems coupling quantum oscillators with coherent states, analyzing entanglement dynamics and decoherence effects.

SKILLS

Programming & Computing: Python (QuTiP, NumPy, SciPy), MATLAB, Fortran, Bash Scripting, Linux System Administration.

Typesetting & Tools: LaTeX, Git/GitHub.

Languages: English (Fluent), Hindi (Native).

EXPERIENCE

Winter School on Recent Frontiers in Physics Research (Online) December 2025
Indian Institute of Science (IISc), Bengaluru

- Participated in advanced lectures covering astrophysics and cosmology, quantum condensed matter, soft and active matter, plasma physics, photonics, and quantum technologies.
- Took part in tutorial sessions involving computational demonstrations and numerical exercises to illustrate physical concepts.
- Engaged with faculty and researchers through interactive discussions, with particular interest in lectures related to astrophysics and cosmology.
- Certificate of Participation

IIA Online Summer School June 2025
Indian Institute of Astrophysics, Bengaluru

- Selected to participate in a summer school hosted by the Indian Institute of Astrophysics.
- Attended lectures and hands-on sessions on advanced topics including galactic structure, observational techniques, radio astronomy, cosmology, and solar physics.
- Engaged with leading scientists through interactive sessions to deepen conceptual and practical understanding of astrophysics.
- Certificate of Completion

Society for Data Science (Member) Feb 2023 – May 2025
BIT, Mesra, Ranchi

- **Upskilled** in data analysis and programming through intensive Python workshops.
- **Spearheaded** logistics and event coordination for society-wide technical initiatives.
- **Secured** event sponsorships by pitching society initiatives to external partners.

CERTIFICATIONS & AWARDS

- **IIT JAM 2025**: Achieved All India Rank (AIR) 220 in the Joint Admission Test for Masters.
- BIT GP Birla Scholarship for Program Topper – Awarded twice (2 semesters) for outstanding academic performance at BIT Mesra